SafeDrop
EWASTE

AWS ENVIRONMENTAL HACKS · WASTE AND ENERGY TRACK

Your old phone still holds your life.
Wipe it. Drop it. Done.

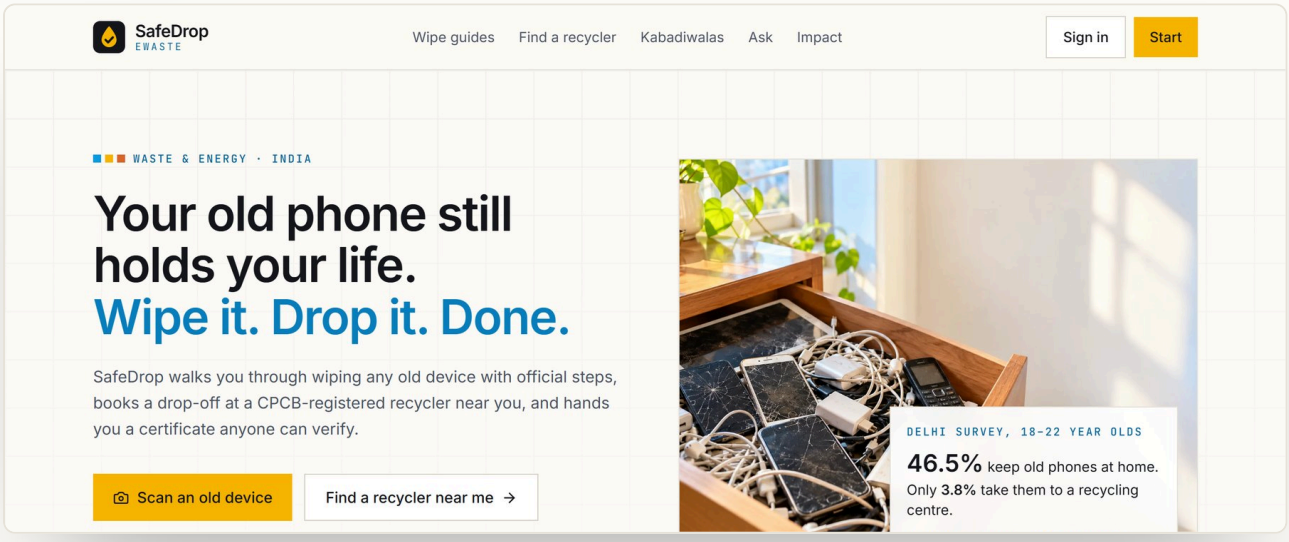
SafeDrop takes an old device from a drawer to a recorded handover at a government-registered recycler: a versioned wipe checklist the owner signs, a single-use signed handover, and a hash-linked receipt that says exactly what it proves.

557
CPCB-listed sites after removing test and disabled rows

8
hash-linked events from drawer to processor confirmation

16 + 16
unit tests and live end-to-end security checks passing

0
made-up numbers: every figure has a source or a measurement



Live app
main.d2ykddnngohyy3.amplifyapp.com

Demo login (every role)
demo@example.com
SafeDrop@2026

Source code
github.com/aditeejha/safedrop-ewaste

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Try it in five minutes. Sign in with `demo@example.com` / `SafeDrop@2026`. The one demo account holds every role: owner, drop-point staff (“Partner”), kabadiwala, recycling facility (“Facility”) and admin. Add a device, answer five condition questions, work through the checklist and sign the attestation. Then book, confirm the handover in Partner, dispatch it, confirm it in Facility, and open the receipt to watch your browser re-verify the hash chain. Demo activity is labelled and never counts towards public totals.

THE PROBLEM

01 Two fears keep devices in drawers

Old electronics pile up at home because people fear for their data and don’t know who will take a device safely. When devices do leave, many go to scrap dealers and into informal yards.

1,397,955 t

E-waste generated in India in FY 2024-25: a CPCB estimate from producers’ sales data and average device life.

PIB · 11 Aug 2025 · India

70.71%

Share reported as collected, dismantled and recycled or disposed formally that year. The rest is unaccounted for.

PIB · 24 Jul 2025 · India

46.5%

Of 1,039 young adults surveyed in New Delhi keep old phones at home; only 3.8% take them to a recycling centre and 11.2% sell to scrap dealers.

Borthakur & Singh · 2022 · New Delhi

26% · 19%

Of people keeping old phones worry about others accessing their data; 19% don’t know how to wipe it. No Indian equivalent was found.

YouGov · May 2024 · UK

Why it matters

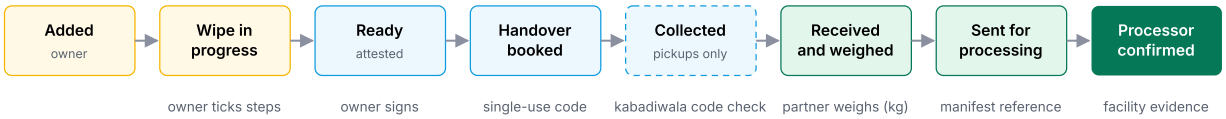
WHO says open burning and heating are the most hazardous informal e-waste activities, linked to stillbirth, premature birth, neurodevelopmental effects from lead and reduced lung function. The last official estimate (Rajya Sabha, 2011) put more than 90% of India’s e-waste in the unorganised sector. The E-Waste (Management) Rules, 2022 require e-waste to reach registered producers, refurbishers or recyclers, and the Battery Waste Management Rules, 2022 ask consumers to discard batteries separately. What is missing is a path a household can follow and trust.

SafeDrop answers **“I don’t know how to erase it”** with a versioned checklist the owner signs, and **“I don’t know who to give it to”** with registered sites, a single-use handover and a receipt that is honest about what it proves.

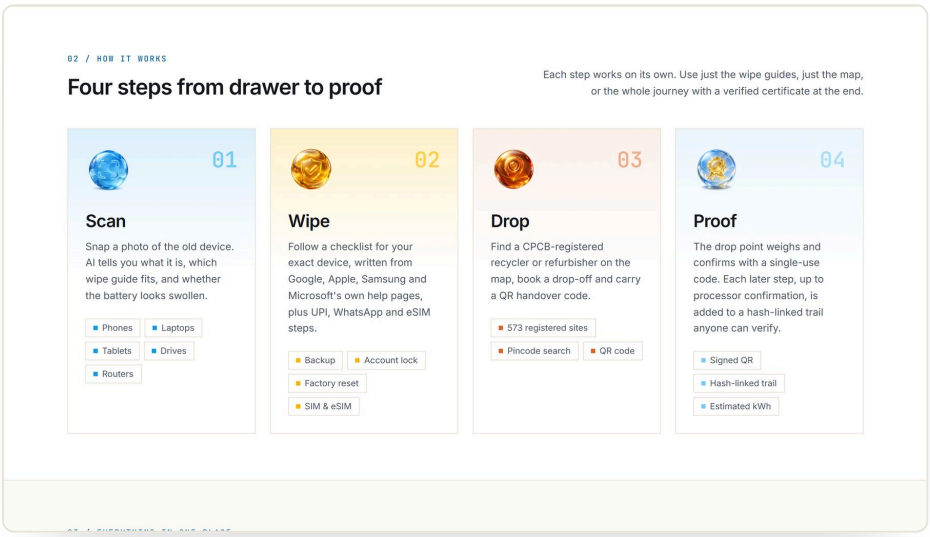
THE JOURNEY

02 One device, one ID, eight recorded steps

Every screen follows the same device ID and the same vocabulary. Only the server can move a device forward, and only along these steps.



Not allowed, and refused by the server: skipping the wipe, booking twice, jumping to “processed”, going backwards, or a partner confirming processing. Each arrow is one transaction that updates the device and appends one hash-linked event (chapter 10).



The home page’s four steps. The wipe guides and the recycler map work without an account; the recorded journey needs one.

THE 15 FEATURES

03 Fewer promises, each one backed by the server

This version reworks the feature list around trust: every feature below is enforced by the backend and covered by a live test (chapter 16), not just shown on a page.

01 **AI device scan**
Gemini vision suggests type, brand and guide; Rekognition is the fallback. Labelled a suggestion, never a diagnosis.

02 **Reuse-first advice**
Five condition questions decide reuse, recycle or "not enough information". No guessing.

03 **Battery hazard stop**
A swollen or leaking battery blocks ordinary drop-off and shows EPA safe-handling steps.

04 **10 versioned wipe guides**
104 steps from makers' own pages; each guide has a version and review date.

05 **Signed wipe record**
Steps are done, doesn't apply or skipped; critical steps can't be skipped; the owner signs.

06 **India privacy steps**
UPI apps, WhatsApp backup, eSIM and Find My Device.

07 **Ask SafeDrop**
Cited answers in English or Hindi, with rule-based buttons for the next safe step.

08 **Cleaned recycler map**
557 CPCB-listed sites with source, check date and location confidence.

09 **Is it registered?**
Name checks show match confidence and say what a match does not prove.

10 **Drop-off or doorstep pickup**
A partner point, or a verified kabadiwala who collects with your code.

11 **Single-use signed QR**
HMAC-signed QR and hashed codes that expire and work once.

12 **Partner portal**
Weigh in kg, correct within 48 h with a reason, dispatch to a registered recycler.

13 **Processor confirmation**
Only the receiving facility, with its own evidence, can mark items processed.

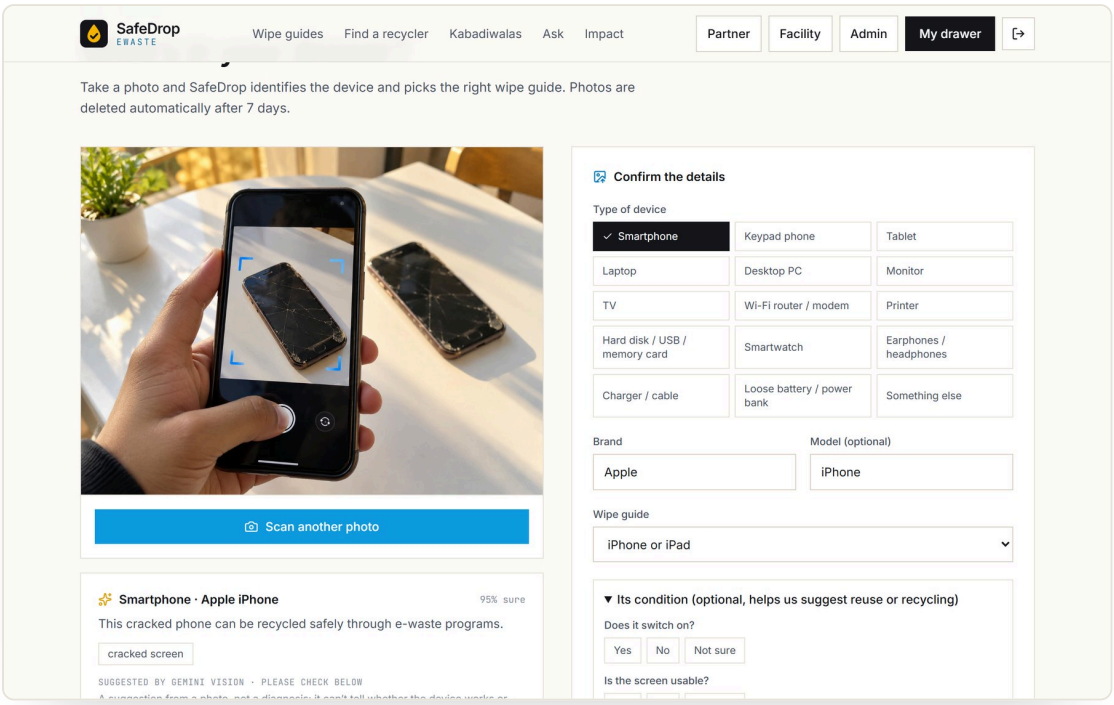
14 **Hash-linked receipt**
Every step is a chained event the browser re-verifies; the receipt lists what it proves.

15 **Evidence-based impact**
Collected, weighed and processed apart; CO_2e and kWh labelled estimates.

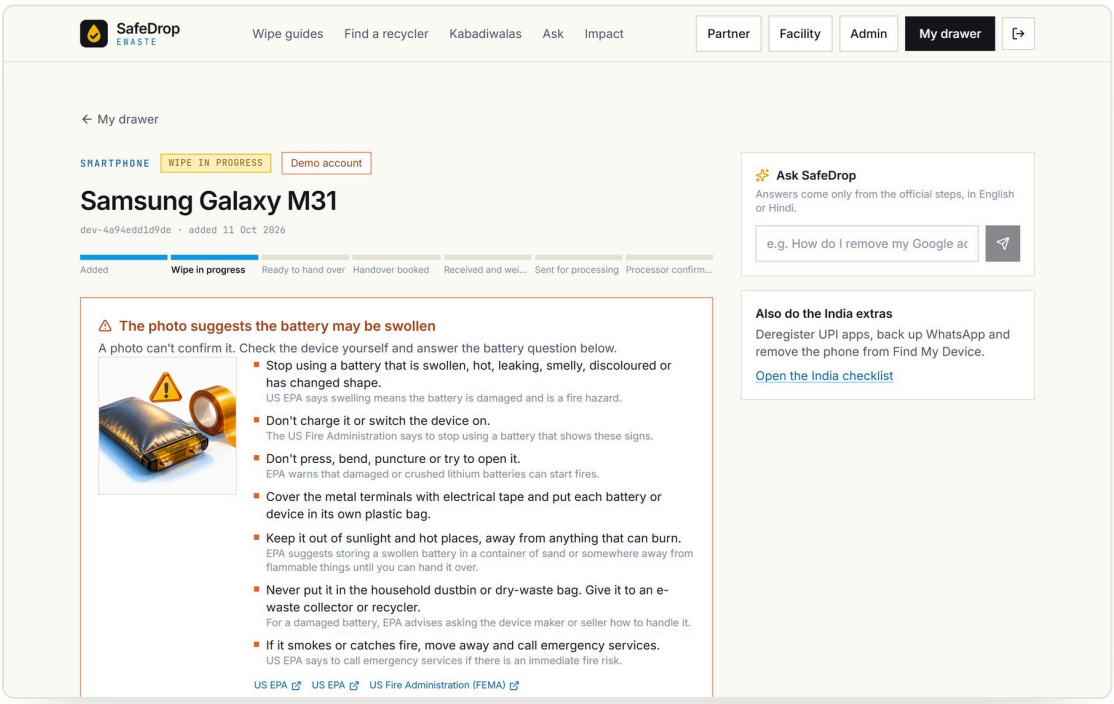
The track asks teams to *"close the loop on what a city throws away and cut what it burns"*. SafeDrop covers four of its listed problems (E-waste, Recycling, Segregation, Informal recyclers) and reports the energy each handover would save once recycled.

WALKTHROUGH

04 Scan, condition and reuse-first



Add a device. The photo is resized in the browser and uploaded with an S3 presigned POST, which lets S3 itself refuse anything over 5 MB or not an image. Gemini suggests the device; the owner corrects it, and the wipe guide follows the brand they type.

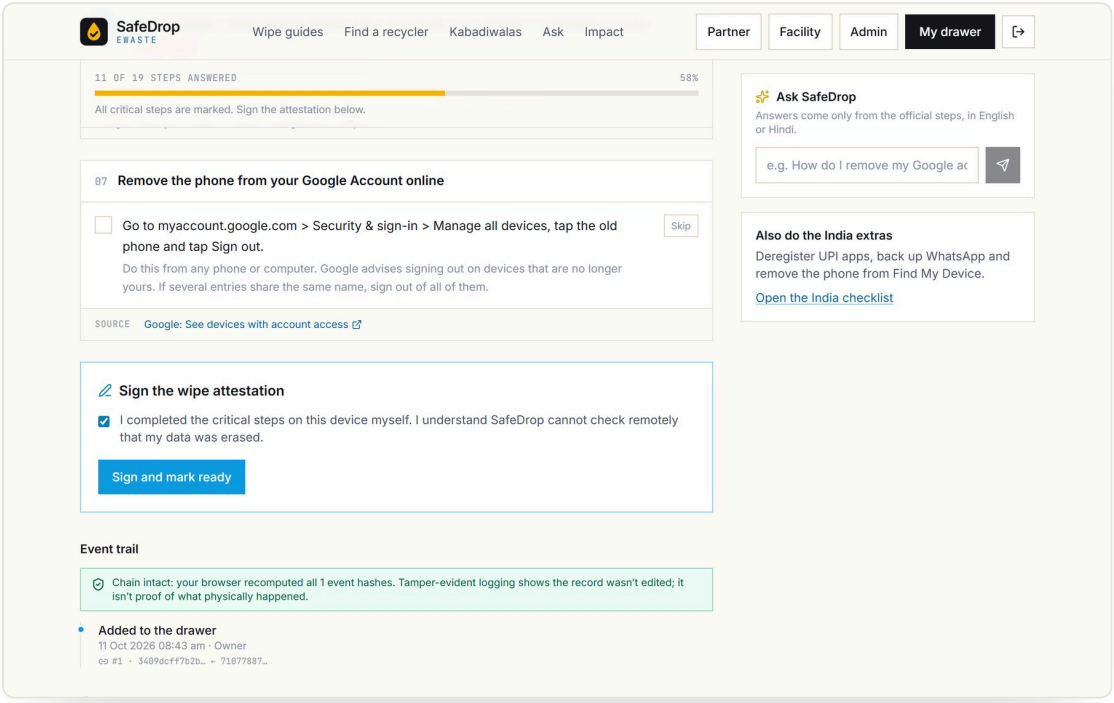


Condition and next step. Answers drive a pure, tested routing rule: a swollen battery means special handling, a dead or badly damaged device means recycling, a working unlocked device means reuse first, and missing answers mean "not enough information".

PRIVACY ASSURANCE

05 Wipe and sign the attestation

A guide alone is not evidence. SafeDrop records which version of which guide the owner followed, how each step ended, and a signed statement that is honest about its limits.

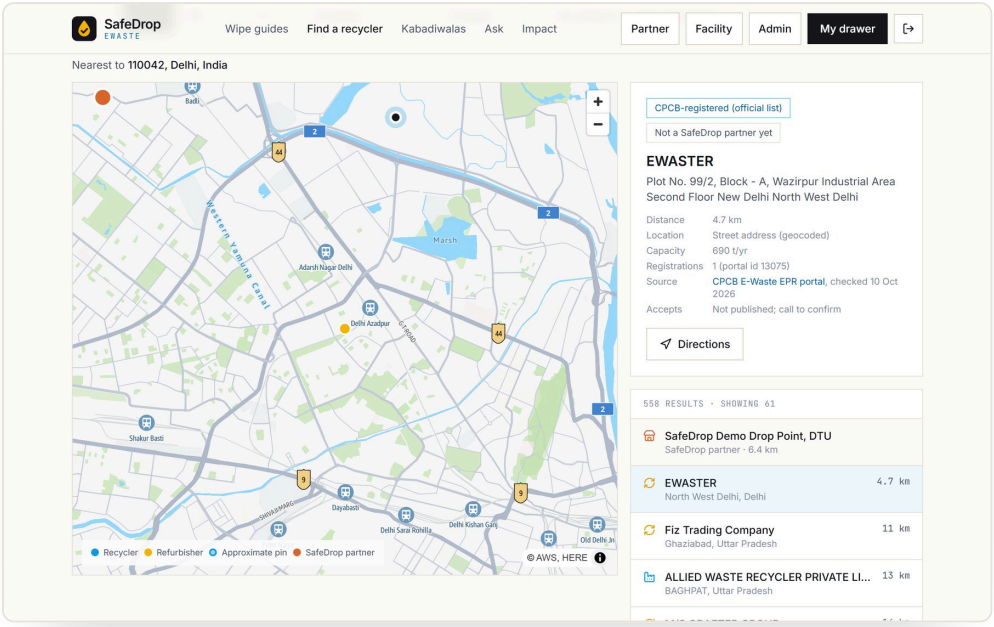


The attestation. It unlocks only when every critical step is done or marked “doesn’t apply”, and the server re-checks that rule. A partial checklist can never show as ready.

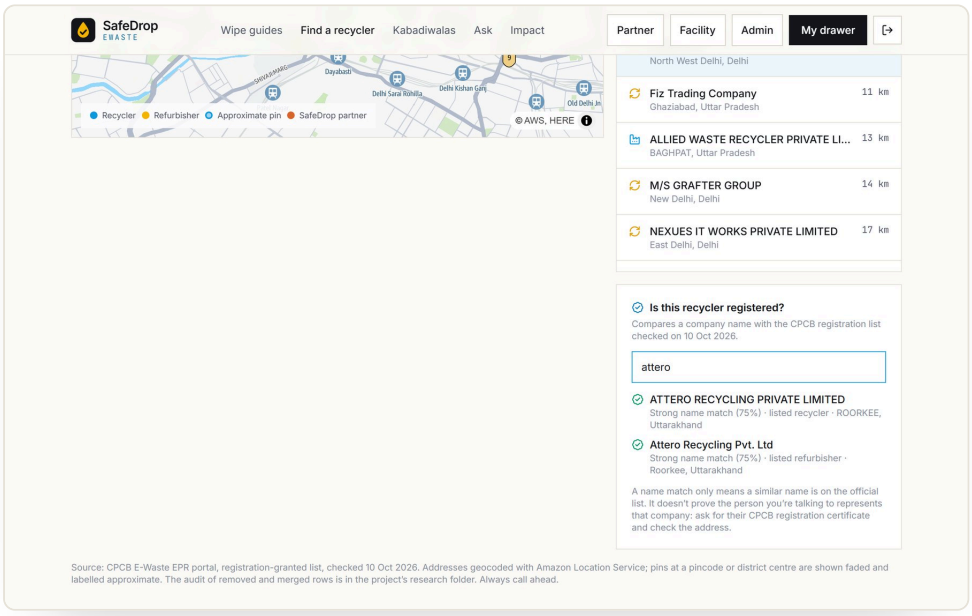
RULE	HOW IT IS ENFORCED
Versioned guides	Each guide’s version is its review date plus a hash of its steps and sources, so the record names the exact text followed.
Three step states	Done, doesn’t apply, or skipped; skipping is refused for critical steps (account locks, eSIM, SIM and memory card, factory reset).
Explicit statement	“I completed the critical steps … SafeDrop cannot check remotely that my data was erased.” Sent as statement version `attest-1`; anything else is rejected.
Never collected	Passwords, PINs, recovery codes, OTPs and account credentials. Photos are private and deleted after 7 days (chapter 13).

RECYCLER TRUST

06 Find a registered site



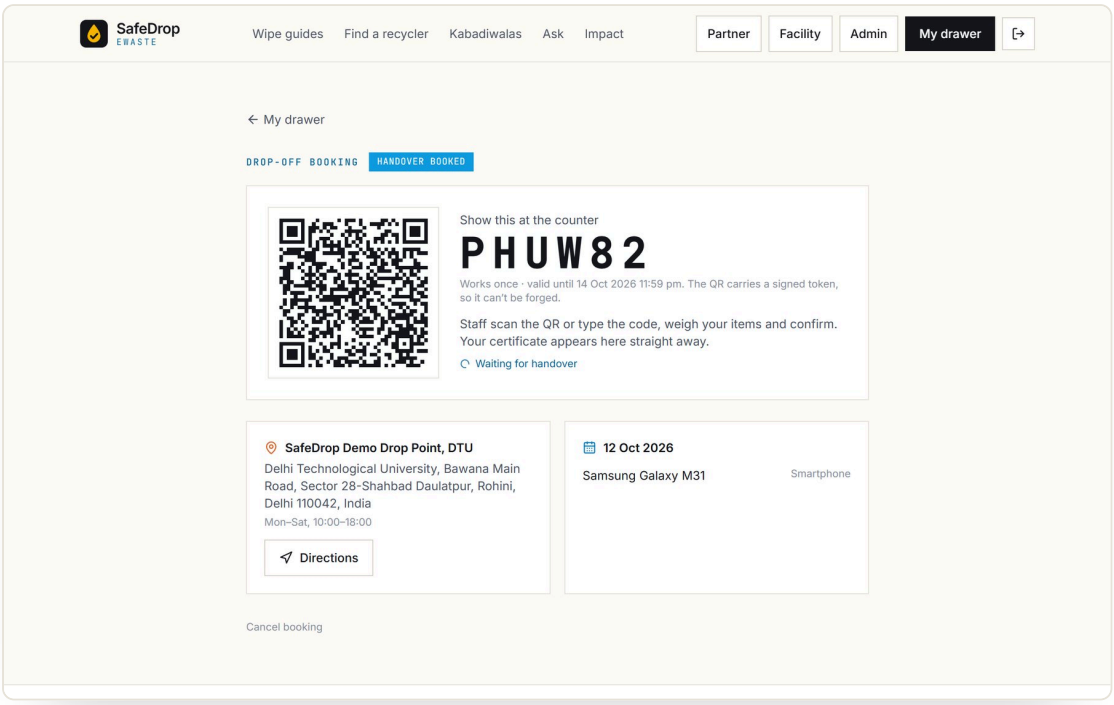
Nearest to pincode 110042. Amazon Location Service renders the map (India political view) and geocodes the pincode. Pins placed at a pincode or district centre are drawn faded and labelled approximate. The card separates “CPCB-registered (official list)” from “SafeDrop partner” and “Confirmed to accept”.



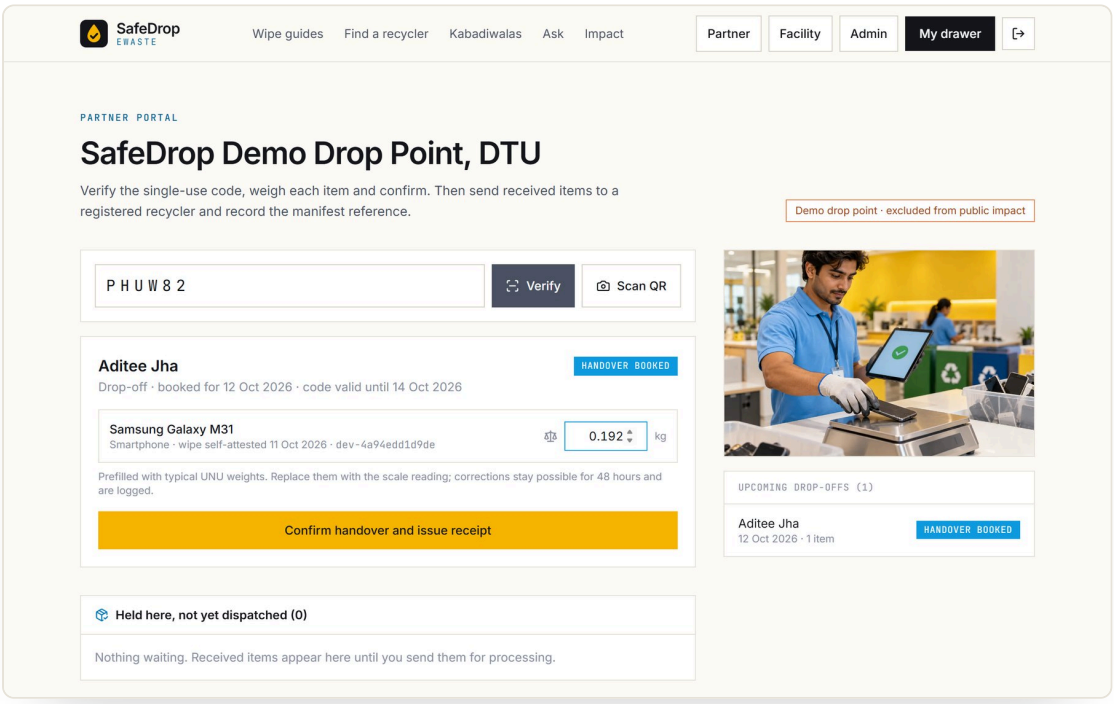
“Is this recycler registered?” Matches are graded exact, strong or partial with a score, ambiguous results are flagged, and the check says plainly that a name match is not proof the person in front of you represents that company.

TRUSTCHAIN

07 Book and hand over



The booking. The QR carries the booking ID and an HMAC signature from a secret in SSM, so it can't be forged. The 6-character code is stored only as a SHA-256 hash, works once, and expires two days after the booked date.

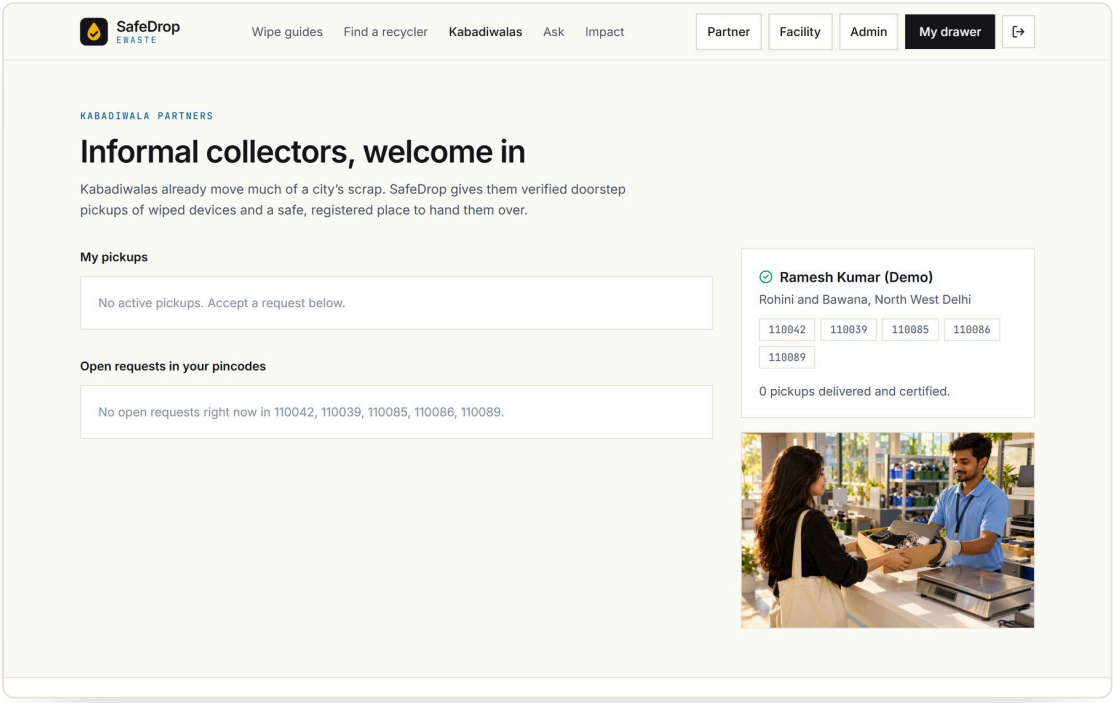


Partner portal. Staff scan or type the code and weigh each item in kg. The server checks the point, the expiry, the booking state, every booked item and the weight range before one transaction confirms it. Every rejection is logged for CloudWatch.

INFORMAL RECYCLERS

08 Kabadiwala pickup partners

Kabadiwalas already move much of a city’s scrap. SafeDrop gives them verified doorstep pickups and a registered place to hand devices over, without pretending they are recyclers.



The pickup partner dashboard. Partners apply with only a name, mobile number and pincodes, give explicit consent to share them, and are approved by an admin after a call. Customers’ addresses are revealed only after a partner accepts.

Chain of custody

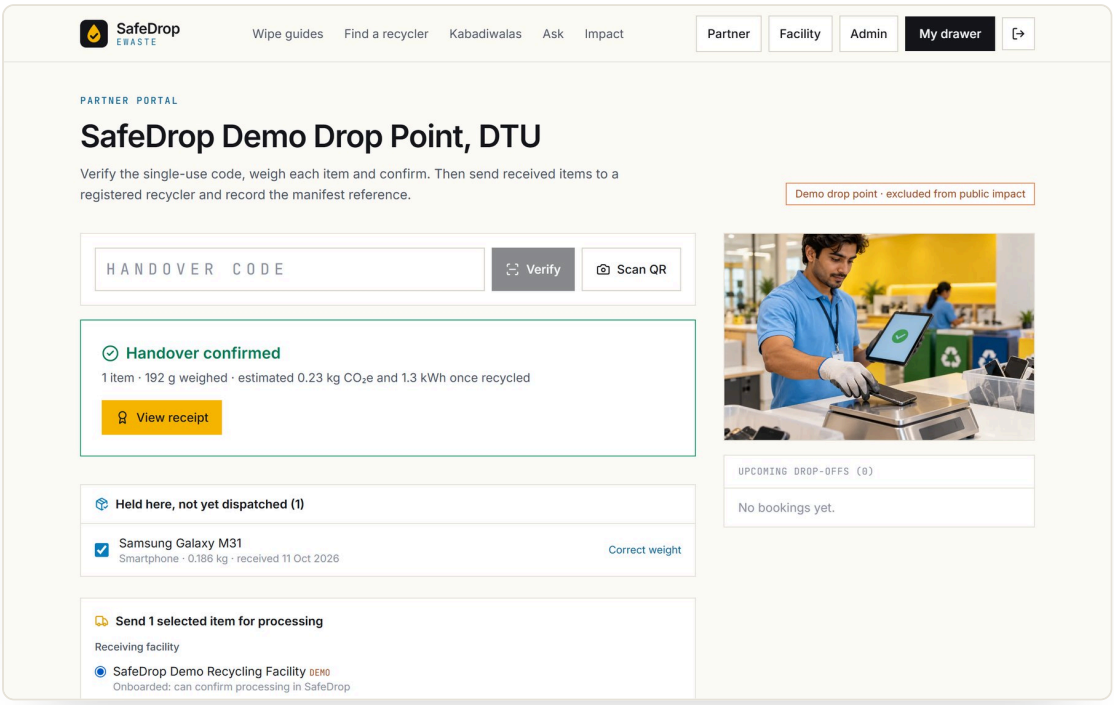
Request → accepted (event `pickup.accepted`) → collected with the customer’s code (`pickup.collected`) → received and weighed at the drop point. The drop point refuses the code until the pickup is collected.

Clear roles

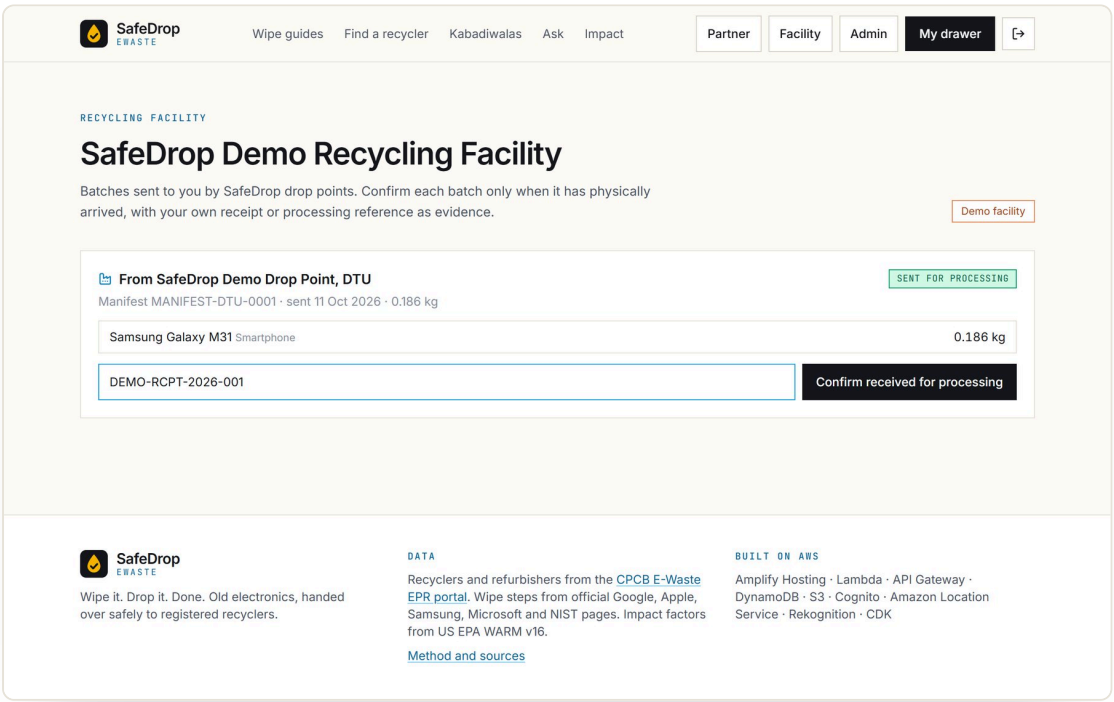
Pickup partners are informal collectors; drop points record weights; only an onboarded facility confirms processing. The receipt names the pickup partner.

BEYOND THE COUNTER

09 Dispatch and processor confirmation



Held here, not yet dispatched. Items received at the point stay listed until sent. Staff can correct a weight within 48 hours with a reason (an audited `weight.corrected` event), then dispatch a batch to a CPCB-registered recycler or an onboarded facility with a manifest reference.

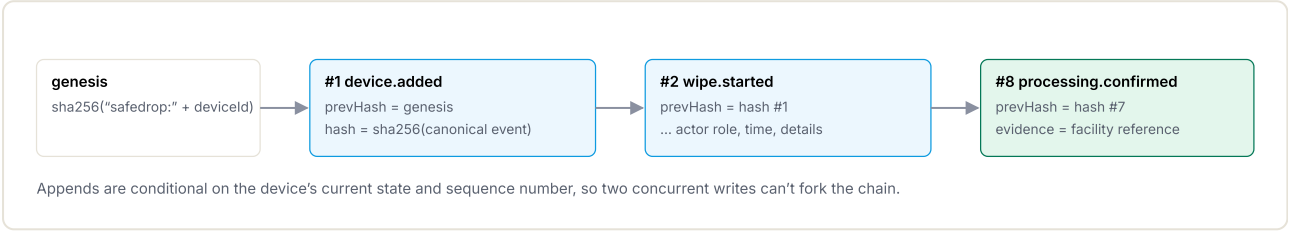


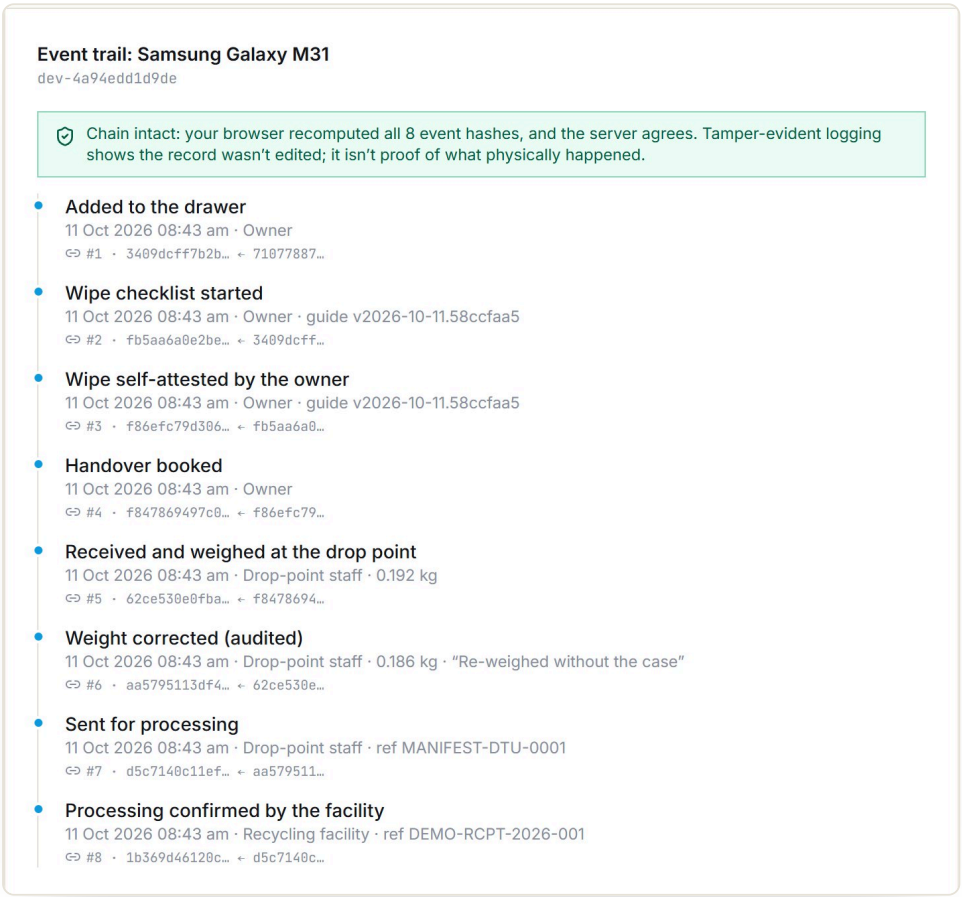
Recycling facility. Only staff linked to the receiving facility can confirm a batch, and only with their own receipt or processing reference. Batches sent to CPCB sites that are not onboarded stay “awaiting confirmation” rather than being assumed processed.

TRUSTCHAIN

10 The hash-linked receipt

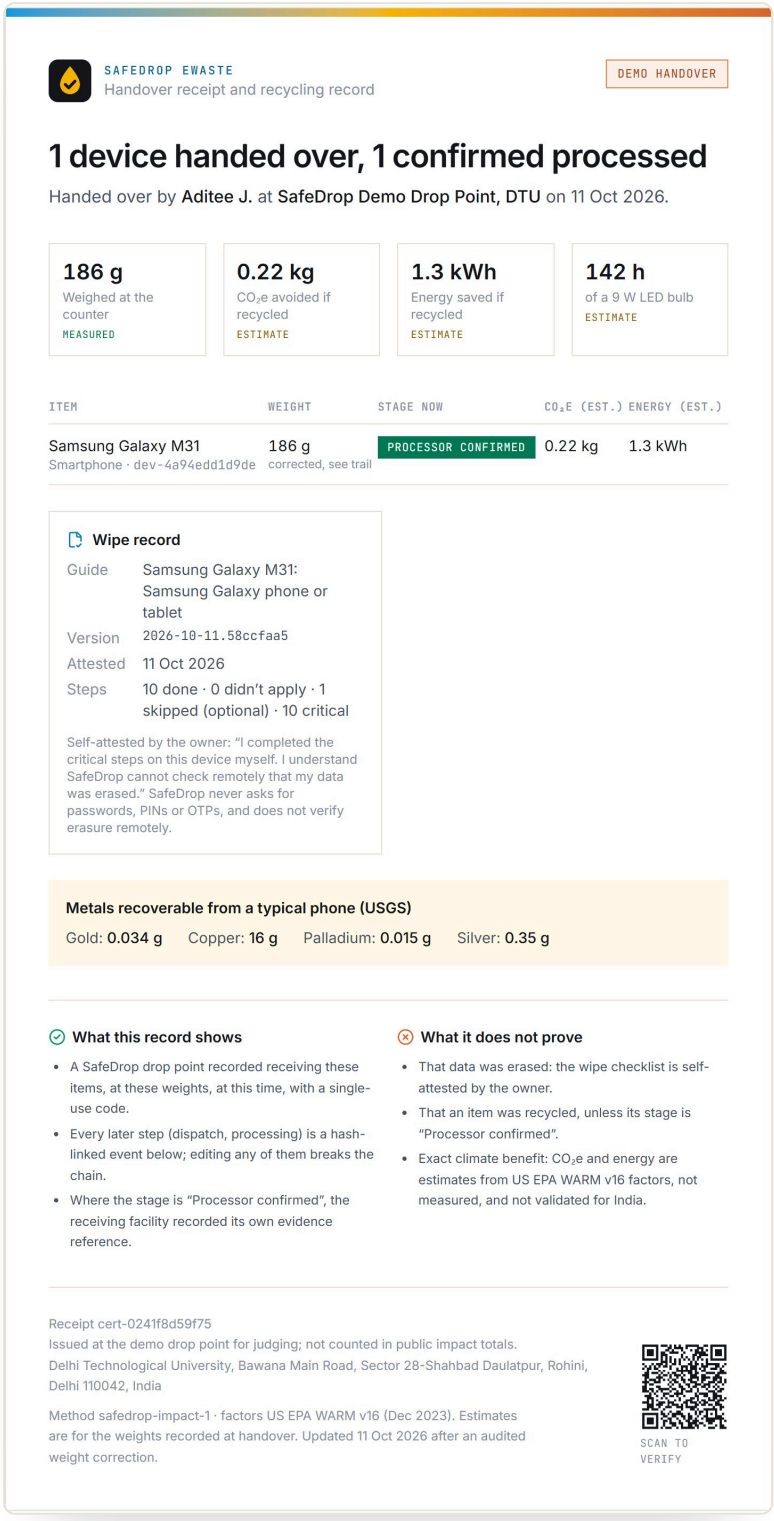
Each device has an append-only trail. Every event stores the hash of the one before it, so editing, deleting or reordering any past event breaks the chain, and anyone can check.





The trail. The browser recomputes all eight SHA-256 hashes with the Web Crypto API and compares with the server's own check. This is tamper-evident logging: it shows the record wasn't edited, not what physically happened.

10 The receipt



SAFEDROP EWASTE
Handover receipt and recycling record

DEMO HANDOVER

1 device handed over, 1 confirmed processed

Handed over by **Aditee J.** at **SafeDrop Demo Drop Point, DTU** on 11 Oct 2026.

186 g	0.22 kg	1.3 kWh	142 h
Weighed at the counter	CO_2e avoided if recycled	Energy saved if recycled	of a 9 W LED bulb
MEASURED	ESTIMATE	ESTIMATE	ESTIMATE

ITEM	WEIGHT	STAGE NOW	CO_2E (EST.)	ENERGY (EST.)
Samsung Galaxy M31 Smartphone · dev-4a94edd1d9de	186 g corrected, see trail	PROCESSOR CONFIRMED	0.22 kg	1.3 kWh

Wipe record

Guide	Samsung Galaxy M31: Samsung Galaxy phone or tablet
Version	2026-10-11.58ccfaa5
Attested	11 Oct 2026
Steps	10 done · 0 didn't apply · 1 skipped (optional) · 10 critical

Self-attested by the owner: "I completed the critical steps on this device myself. I understand SafeDrop cannot check remotely that my data was erased." SafeDrop never asks for passwords, PINs or OTPs, and does not verify erasure remotely.

Metals recoverable from a typical phone (USGS)

Gold: **0.034 g** Copper: **16 g** Palladium: **0.015 g** Silver: **0.35 g**

What this record shows

- A SafeDrop drop point recorded receiving these items, at these weights, at this time, with a single-use code.
- Every later step (dispatch, processing) is a hash-linked event below; editing any of them breaks the chain.
- Where the stage is "Processor confirmed", the receiving facility recorded its own evidence reference.

What it does not prove

- That data was erased: the wipe checklist is self-attested by the owner.
- That an item was recycled, unless its stage is "Processor confirmed".
- Exact climate benefit: CO_2e and energy are estimates from US EPA WARM v16 factors, not measured, and not validated for India.

Receipt cert-0241f8d59f75
Issued at the demo drop point for judging; not counted in public impact totals.
Delhi Technological University, Bawana Main Road, Sector 28-Shahbad Daulatpur, Rohini, Delhi 110042, India

Method safedrop-impact-1 · factors US EPA WARM v16 (Dec 2023). Estimates are for the weights recorded at handover. Updated 11 Oct 2026 after an audited weight correction.

SCAN TO VERIFY

The receipt. Measured weight and live stage per device, the signed wipe record, estimates labelled as such, and a "what this shows / what it does not prove" panel. It is built only from server records; nothing on it is typed by the user.

AI

11 Ask SafeDrop: grounded answers and safe actions



A battery question. The answer cites US EPA and US Fire Administration pages, with citation numbers matched to the listed sources, and offers buttons for the next step.

- **Grounded:** the model sees only the relevant official guide steps, the India extras and the battery-safety steps, and is told to say when the sources don't answer.
- **Actions by rules, not by the model:** buttons such as "Open this guide", "Show registered recyclers", "Prepare a pickup request" and "Battery safety steps" are chosen by tested keyword rules. They only open pages; nothing changes until the user confirms there.
- **Resilient and bounded:** four Gemini Flash models with cooldowns, a guide-search fallback, and 40 AI calls per user or IP per day. Amazon Bedrock Nova can be switched on when the account allows it.

IMPACT LEDGER

12 Waste to energy, counted honestly

Recovered metals replace mining and refining. SafeDrop counts that saving only from weights measured at a drop point, and keeps collected, weighed and processor-confirmed apart.

```
CO₂e avoided (kg, estimate) = kg ÷ 907.18 × (landfilling - recycling) [WARM v16, t CO₂e / short ton] × 1000
Energy saved (kWh, estimate) = kg ÷ 907.18 × (landfilling - recycling) [WARM v16, million Btu / short ton] × 293.07
Formula version safedrop-impact-1 · factors US EPA WARM v16 (Dec 2023)
```

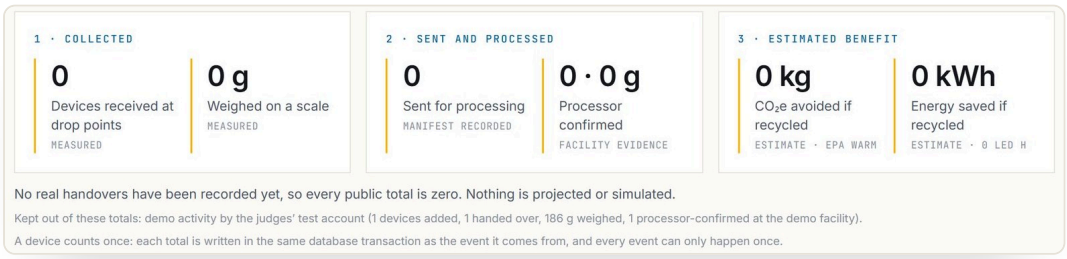
WARM MATERIAL	SAFEDROP DEVICE TYPES	CO_2E T / SHORT TON	KWH / SHORT TON
Portable Electronic Devices	Phones, keypad phones, tablets, laptops, smartwatches	1.08	6,219
Desktop CPUs	Desktop PCs	1.51	6,295
Flat-Panel Displays	Monitors, TVs	1.01	4,496
Hard-Copy Devices	Printers	0.58	2,397
Mixed Electronics	Routers, drives, earphones, chargers, other	0.92	4,188

Counted once

One ledger row per weigh event, written in the same DynamoDB transaction as the event, with a condition that it doesn't already exist. A replayed or retried handover can't add impact. Corrections supersede the old row and adjust totals by the difference.

Labelled and limited

"Device added", "booking created" and "QR scanned" never count as recycling. CO_2e and kWh are shown as estimates: WARM describes US conditions and is not validated for India. Demo activity has its own counter.

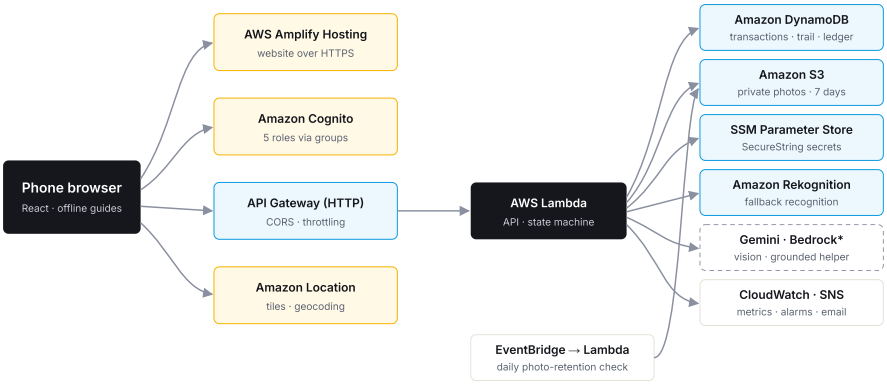


The public ledger. No real handovers yet, so every public total is zero; demo activity is listed separately.

ARCHITECTURE

13 How SafeDrop uses AWS

Each AWS service has one job: Amplify → API Gateway → Lambda → DynamoDB at the core, with Cognito, S3, Amazon Location and CloudWatch from day one.

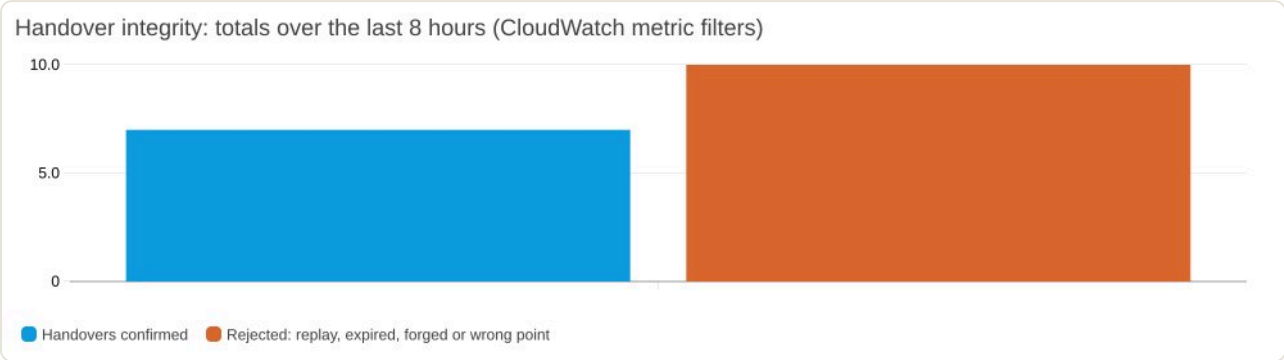


One AWS CDK stack (infra/lib/stack.ts) in Mumbai (ap-south-1). * Bedrock is wired in behind a flag, pending AWS account verification.

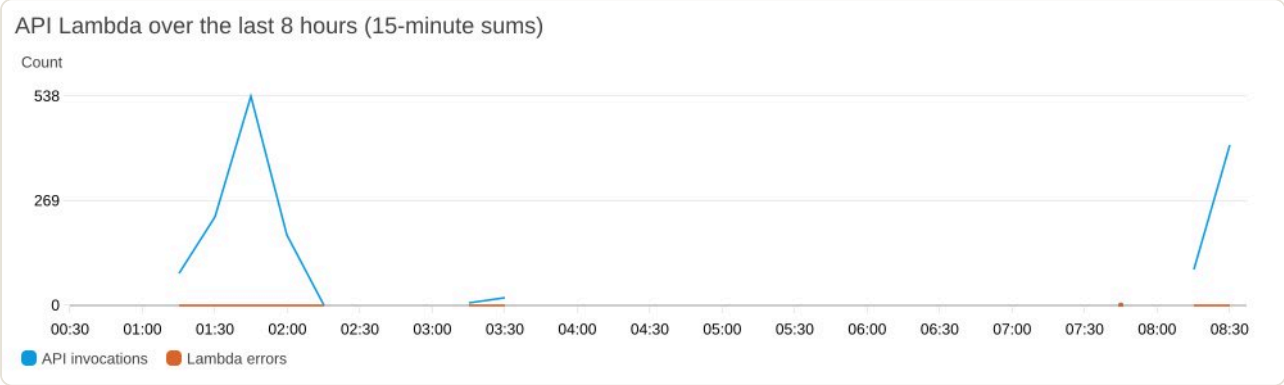
AWS SERVICE	ITS ONE JOB	GUARDRAIL
Lambda + API Gateway	All business rules: lifecycle, handovers, dispatch, ledger, AI calls	Schema, role and ownership checks in every handler; stage throttling; preflights before auth
DynamoDB	Devices, trails, bookings, receipts, ledger, partners	Conditional writes and transactions for single-use codes and exactly-once impact; idempotency rows with TTL
Cognito	Sign-in; groups for partner, processor, admin and demo	Groups checked server-side; web client can't write the `pointId` attribute
S3	Private device photos	Public access blocked, encryption, presigned POST with 5 MB limit, 7-day lifecycle, Lambda limited to `photos/*`
SSM Parameter Store	Gemini key and QR signing secret	SecureString, written by the deploy script via the SDK; Lambda can read only these two parameters
Amazon Location	Map tiles and geocoding	Key restricted to the site's domains; approximate pins labelled; never used as proof of registration
EventBridge + Lambda	Daily photo-retention check	Counts photos older than 8 days, deletes them and publishes metrics
CloudWatch + SNS	Structured logs, metric filters, dashboard, alarms by email	Logs carry event names and counts, never names, codes or tokens
Amplify, Budgets, CDK	Hosting, a $5 monthly alert, infrastructure as code	Every script refuses to run against any account but the hackathon one

ARCHITECTURE, CONTINUED

13 Observability, and what we left out on purpose



Live CloudWatch metric filters, rendered by the CloudWatch API: confirmed handovers against refused attempts (replays, expired, forged and wrong-point codes from the security tests). An alarm emails when 10 or more are refused within 15 minutes.



The API Lambda over 8 hours. Zero Lambda errors and zero 5xx responses in that window; the peaks are test and screenshot runs.

Proven, not promised

The photo-retention job was run on the live stack: 9 photos checked, 0 older than 8 days. The S3 lifecycle rule (`photos/` , 7 days) is enabled, and four alarms (replay attempts, API errors, stale photos, retention-job errors) report OK.

Deliberately not added

SQS, Step Functions, DynamoDB Streams: no long-running or retried job exists yet, and impact is written in the same transaction as its event. **Secrets Manager:** SSM SecureString does the job at no cost. **CloudFront:** blocked until the account is verified, so Amplify Hosting serves the site.

DESIGN CHOICES

14 Data model, privacy and security

RECORD	KEY	NOTES
Device	`USER#id` / `DEVICE#id`	State, trail head (seq + hash), wipe record, condition; `HELD#point` index while at a drop point
Trail event	`TRAIL#device` / `EV#00001`	Append-only, hash-linked
Booking	`BOOKING#id` / `META`	Indexed by owner, and by drop point, pickup pincode or collector
Handover code	`CODE#sha256(code)`	Created only if absent; deleted on handover or cancel
Receipt, dispatch	`CERT#id` , `DISPATCH#id`	Built from server records only
Impact entry	`IMPACT#device` / `EV#seq`	Measured kg, factors and formula version, status

Security checks the live tests prove

Isolation and roles

User B gets 404 for user A's devices and bookings. Unauthenticated partner calls get 401; non-partners, non-processors and non-admins get 403.

Single-use handovers

Forged QR → 400. Replay → 409. Expired → 410. Two simultaneous scans → exactly one 201. Negative weights and non-kg units → 400.

No shortcuts

Partial checklists can't be attested; critical steps can't be skipped; a partner can't mark items processed; nothing can be dispatched twice.

Retries are safe

An Idempotency-Key replays the stored response, so a flaky network can't create a second device, booking or handover.

REAL DATA ONLY

15 Real data and the directory audit

DATASET	SOURCE	SIZE	USED FOR
Registered sites after the audit below	CPCB E-Waste EPR portal, registration-granted lists, checked 10 Oct 2026	557	Map, nearest search, “Is it registered?”, dispatch targets
Locations	Amazon Location Service geocoding with precision recorded	152 approx.	Faded pins and “approximate” labels
Wipe guides	Google, Samsung, Apple, Microsoft, WhatsApp, HMD, TP-Link, HP, NIST SP 800-88 Rev. 2	104 steps	Versioned checklists; the AI helper’s only knowledge
Impact factors	US EPA WARM v16 (Dec 2023): Exhibits 1-5 and 7-12	7 materials	Estimated CO_2e and kWh
Context facts	PIB/CPCB, Global E-waste Monitor 2024, WHO, surveys, E-Waste and Battery Rules 2022	45	Home and impact pages

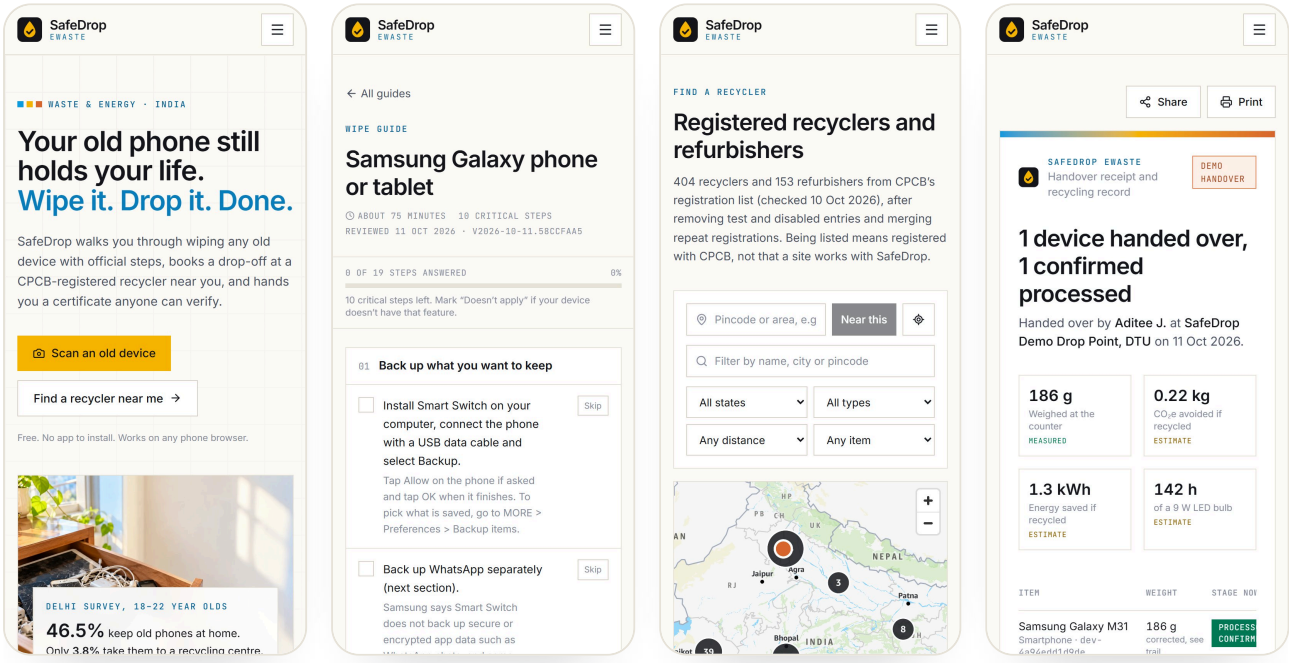
What the audit changed

FINDING	ACTION
2 test rows in CPCB’s own list	“Recycler Test” (Delhi) and “Recycler Test 2” (Agra) with placeholder addresses: removed and logged
6 rows marked “operations disabled”	Removed and logged
Repeat registrations of one site	5 sites with 2–5 registrations at the same name and pincode merged into one listing that shows every portal ID
Same name, same district, different pincode	2 cases kept separate and flagged for human review in `research/directory-audit.md`

TESTING

16 Testing and validation

TEST	WHAT IT COVERS	RESULT
Unit tests `npm test`	Lifecycle transitions, hash chain and tamper detection, QR signatures and code hashing, reuse-first routing, helper actions, JSON parsing, impact maths, citations, labels	`16 / 16 PASS`
Live end-to-end and security `scripts/e2e.mjs`	Against the deployed stack with a second throwaway user: isolation, roles, partial wipe, idempotent retry, hazard stop, forged and expired codes, replay, simultaneous double scan, correction, dispatch, processor confirmation, 8-event chain, kabadiwala pickup, S3 5 MB limit, demo exclusion	`16 / 16 PASS`
Browser flows	Edge driving the live site at 1366 px and 390 px: sign in, add, tick, sign attestation, book, confirm handover, open receipt and re-verify the chain in the browser	`PASS, NO PAGE ERRORS`
Page sweep	16 pages at two widths, including Partner, Facility, Admin and Battery safety: console errors and sideways overflow	`32 / 32 CLEAN`



On a 390 px phone: home, Samsung guide (works offline once opened), recycler map and receipt.

THIS VERSION

17 What changed in existing features

A quality pass on everything that already existed: each claim made traceable, each status made truthful.

AREA	BEFORE	NOW
Home stats	Source shown, geography implicit	Definition, date and geography beside each figure; UK survey labelled UK; live totals start at zero
Recycler map	All portal rows, precision in the card	Test, disabled and repeat rows handled; three trust states; faded approximate pins; distance and acceptance filters; empty states
Registered check	Name match shown as registered	Graded match with score, source and date; ambiguity flagged; says what a match doesn't prove
Wipe guides	Unversioned; done only	Version and review date; done, doesn't apply, skipped; works offline
My Drawer	"Mark as wiped"	Signed self-attestation; "Wipe self-attested" wording; live stage and trail on every device
Certificates	"Verified", kg and estimates	"Recorded handover" receipt with stage per device, wipe record, chain check and proves / doesn't prove
Impact page	Single counter set	Collected, weighed, processed apart; measured vs estimated; written exactly once
Photo upload	Presigned PUT	Presigned POST with size and type limits; S3 access narrowed; deletion proven daily
Partner portal	Code, weigh, confirm	Signed QR, expiry, replay logging, audited corrections, dispatch to a recycler
Ask SafeDrop	Answers only	Battery safety included; renumbered citations; next-step buttons
Kabadiwala onboarding	Name, phone, pincodes	Requirements explained, explicit consent, admin approval, informal vs registered made clear

HONEST LIMITS

18 Limits and next steps

TODAY	NEXT
One demo drop point and one demo facility run the partner and processor flows; CPCB sites are listed but not onboarded.	Let registered recyclers claim their listing with registration details, publish what they accept, and confirm processing themselves.
The wipe is self-attested; SafeDrop cannot verify erasure.	Partner-side spot checks recorded as events, and vendor-provided erase reports where available.
Impact uses US EPA WARM factors, not validated for India.	Use Indian recycler process data once partners share it.
CloudFront and Bedrock are blocked until AWS verifies the new account.	Switch Bedrock Nova on with `BEDROCK_ENABLED=true`; move hosting behind CloudFront.
English interface; Hindi answers from the helper.	Hindi and other Indian languages for the interface and guides; offline queue for partners.

What works today: a household can wipe a device with a versioned checklist and sign for it, find a registered site, hand it over through a partner point or a verified kabadiwala, and get a receipt whose every later step is a hash-linked event anyone can verify. All of it runs on AWS, from real data.

REFERENCES

19 Sources

Screenshots and CloudWatch charts are from the live deployment on 11 October 2026. Product images on the website are AI-generated illustrations, labelled as such where they depict real-world scenes.

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